

Chi-Square Tests

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graph TD; A[Chi-Square Tests] --> B[Test of Multinomial Experiment]; A --> C[Test for independence];
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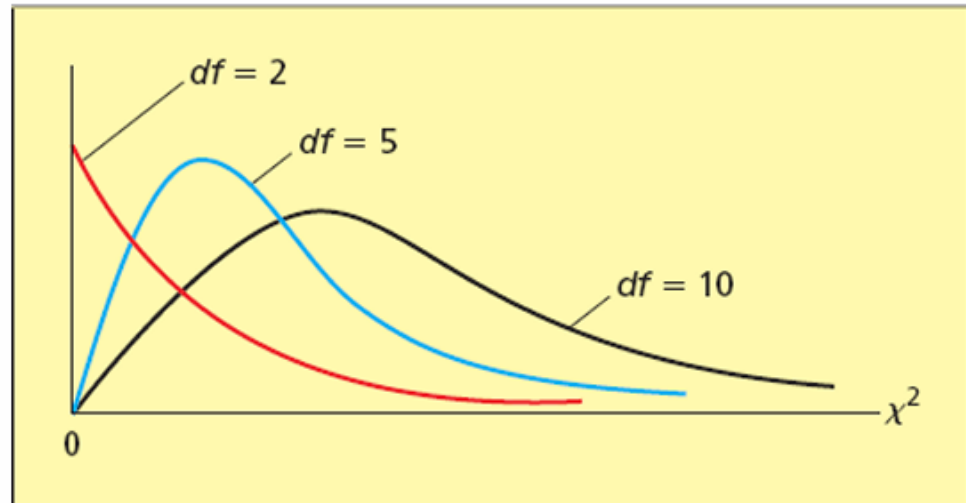
Test of Multinomial Experiment

Test for independence

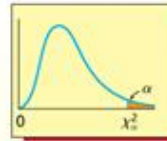
The Chi-Square Distribution

The chi-square χ^2 distribution depends on the number of degrees of freedom.

- There is a family of chi-square distributions.
- It is skewed to the right.
- It is non-negative.

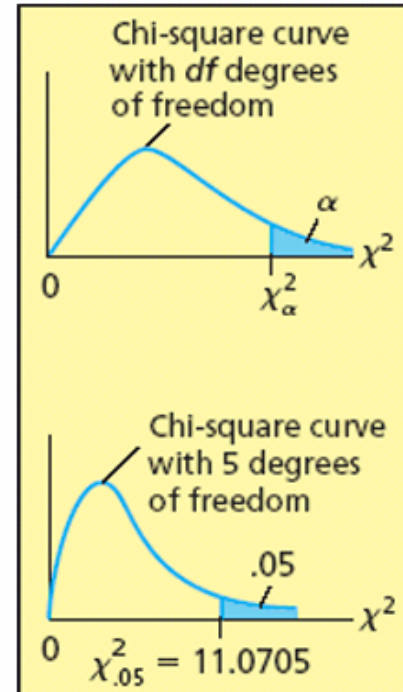


Chi-Square Table



df	$\chi^2_{.995}$	$\chi^2_{.99}$	$\chi^2_{.975}$	$\chi^2_{.95}$	$\chi^2_{.90}$	$\chi^2_{.10}$	$\chi^2_{.05}$	$\chi^2_{.025}$	$\chi^2_{.01}$	$\chi^2_{.005}$
1	.0000393	.0001571	.0009821	.0039321	.0157908	2.70554	3.84146	5.02389	6.63490	7.87944
2	.0100251	.0201007	.0506356	.102587	.210720	4.60517	5.99147	7.37776	9.21034	10.5966
3	.0717212	.114832	.215795	.341846	.584375	6.25139	7.81473	9.34840	11.3449	12.8381
4	.206990	.297110	.484419	.710721	1.063623	7.77944	9.48773	11.1433	13.2767	14.8602
5	.411740	.554300	.831211	1.145476	1.61031	9.23635	11.0705	12.8325	15.0863	16.7496
6	.675727	.872085	1.237347	1.63539	2.20413	10.6446	12.5916	14.4494	16.8119	18.5476
7	.989265	1.239043	1.68987	2.16735	2.83311	12.0170	14.0671	16.0128	18.4753	20.2777
8	1.344419	1.646482	2.17973	2.73264	3.48954	13.3616	15.5073	17.5346	20.0902	21.9550
9	1.734926	2.087912	2.70039	3.32511	4.16816	14.6837	16.9190	19.0228	21.6660	23.5893
10	2.15585	2.55821	3.24697	3.94030	4.86518	15.9871	18.3070	20.4831	23.2093	25.1882
11	2.60321	3.05347	3.81575	4.57481	5.57779	17.2750	19.6751	21.9200	24.7250	26.7569
12	3.07382	3.57056	4.40379	5.22603	6.30380	18.5494	21.0261	23.3367	26.2170	28.2995
13	3.56503	4.10691	5.00874	5.89186	7.04150	19.8119	22.3621	24.7356	27.6883	29.8194
14	4.07468	4.66043	5.62872	6.57063	7.78953	21.0642	23.6848	26.1190	29.1413	31.3193
15	4.60094	5.22935	6.26214	7.26094	8.54675	22.3072	24.9958	27.4884	30.5779	32.8013
16	5.14224	5.81221	6.90766	7.96164	9.31223	23.5418	26.2962	28.8454	31.9999	34.2672
17	5.69724	6.40776	7.56418	8.67176	10.0852	24.7690	27.5871	30.1910	33.4087	35.7185
18	6.26481	7.01491	8.23075	9.39046	10.8649	25.9894	28.8693	31.5264	34.8053	37.1564
19	6.84398	7.63273	8.90655	10.1170	11.6509	27.2036	30.1435	32.8523	36.1908	38.5822
20	7.43386	8.26040	9.59083	10.8508	12.4426	28.4120	31.4104	34.1696	37.5662	39.9968
21	8.03366	8.89720	10.28293	11.5913	13.2396	29.6151	32.6705	35.4789	38.9321	41.4010
22	8.64272	9.54249	10.9823	12.3380	14.0415	30.8133	33.9244	36.7807	40.2894	42.7956
23	9.26042	10.19567	11.6885	13.0905	14.8479	32.0069	35.1725	38.0757	41.6384	44.1813
24	9.88623	10.8564	12.4011	13.8484	15.6587	33.1963	36.4151	39.3641	42.9798	45.5585
25	10.5197	11.5240	13.1197	14.6114	16.4734	34.3816	37.6525	40.6465	44.3141	46.9278
26	11.1603	12.1981	13.8439	15.3791	17.2919	35.5631	38.8852	41.9232	45.6417	48.2899
27	11.8076	12.8786	14.5733	16.1513	18.1138	36.7412	40.1133	43.1944	46.9630	49.6449
28	12.4613	13.5648	15.3079	16.9279	18.9392	37.9159	41.3372	44.4607	48.2782	50.9933
29	13.1211	14.2565	16.0471	17.7083	19.7677	39.0875	42.5569	45.7222	49.5879	52.3356
30	13.7867	14.9535	16.7908	18.4926	20.5992	40.2560	43.7729	46.9792	50.8922	53.6720
40	20.7065	22.1643	24.4331	26.5093	29.0505	51.8050	55.7585	59.3417	63.6907	66.7659
50	27.9907	29.7067	32.3574	34.7642	37.6886	63.1671	67.5048	71.4202	76.1539	79.4900
60	35.5346	37.4848	40.4817	43.1879	46.4589	74.3970	79.0819	83.2976	88.3794	91.9517
70	43.2752	45.4418	48.7576	51.7393	55.3290	85.5271	90.5312	95.0231	100.425	104.215
80	51.1720	53.5400	57.1532	60.3915	64.2778	96.5782	101.879	106.629	112.329	116.321
90	59.1963	61.7541	65.6466	69.1260	73.2912	107.565	113.145	118.136	124.116	128.299
100	67.3276	70.0648	74.2219	77.9295	82.3581	118.498	124.342	129.561	135.807	140.169

- A chi-square point χ^2_{α} is the point under a chi-square distribution that gives right-hand tail area α .
- Values of χ^2_{α} are tabulated in a chi-square table
- For example,
If $df=5$ and $\alpha=0.05$
from the table we find that
 $\chi^2_{0.05} = 11.0705$.



A Goodness of Fit Test for Multinomial Probabilities

Binomial vs. Multinomial Experiments

- Carry out n identical trials with k possible outcomes of each trial
- Probabilities are denoted $p_1, p_2, \dots, p_k$ where $p_1 + p_2 + \dots + p_k = 1$
- The trials are independent
- The results are observed frequencies of the number of trials that result in each of k possible outcomes, denoted $f_1, f_2, \dots, f_k$

- Step1. H0: multinomial probabilities are $p_1, p_2, \dots, p_k$

Ha: at least one of the probabilities differs from $p_1, p_2, \dots, p_k$

- Step2 Choose α
- Step3. Multinomial Experiment

Assumption: The expected cell frequencies are ≥ 5

- Step4. Critical value is χ^2_{α} , with $k-1$ degrees of freedom

- Step5
$$\chi^2 = \sum_{i=1}^k \frac{(f_i - E_i)^2}{E_i}$$

- Step6. Reject H0 if $\chi^2 > \chi^2_{\alpha}$

Are technical support calls equal across all days of the week? (i.e., do calls follow a uniform distribution?)

Test with $\alpha = .05$

- Sample data for 10 days per day of week:

	<u>Sum of calls for this day:</u>
Monday	290
Tuesday	250
Wednesday	238
Thursday	257
Friday	265
Saturday	230
Sunday	192
	<u>Total= 1722</u>

Step 1: H_0 : The distribution of calls is uniform over days of the week

$$(p_1=p_2=p_3=p_4=p_5=p_6=p_7=1/7)$$

H_a : The distribution of calls is not uniform

(At least one of p_i is not $1/7$)

Step 2: The level of significance is set at .05.

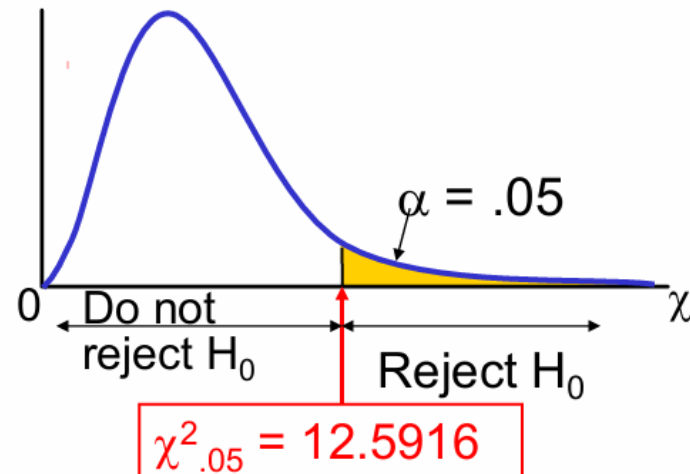
Step 3: It is a multinomial experiment with 7 outcomes.

Step 4: The degrees of freedom equal $7-1=6$.
The critical χ^2 at 6d.f. is 12.5916.

	Observed f_i	Expected E_i	$\frac{(f_i - E_i)^2}{E_i}$
Monday	290	246	$(290-246)^2/246$
Tuesday	250	246	$(250-246)^2/246$
Wednesday	238	246	$(238-246)^2/246$
Thursday	257	246	$(257-246)^2/246$
Friday	265	246	$(265-246)^2/246$
Saturday	230	246	$(230-246)^2/246$
Sunday	192	246	$(192-246)^2/246$
TOTAL	1722	1722	$\chi^2=23.05$

Reject H_0 if $\chi^2 > \chi^2_\alpha$

(with $k - 1$ degrees of freedom)



Conclusion:

$$\chi^2 = 23.05 > 12.5916 = \chi^2_\alpha$$

So **reject H_0** and conclude that the distribution is not uniform

A chi-square test for independence

Step1. H_0 : X and Y are independent

H_a : X and Y are dependent

$$\hat{E}_{ij} = \frac{r_i c_j}{n}$$

Step3. Assumption: The expected cell frequencies are ≥ 5

Step4. Critical value is χ_{α}^2 , with
degrees of freedom $(r-1)(c-1)$

Step5. Test statistic:

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(f_{ij} - E_{ij})^2}{E_{ij}}$$

where:

f_{ij} = observed frequency in cell (i, j)

E_{ij} = expected frequency in cell (i, j)

r = number of rows

c = number of columns

Step6. Reject H_0 if $\chi^2 > \chi_{\alpha}^2$

A sample of 150 accidents were classified by type and gender.

<i>Observed frequencies (f_o)</i>			
<i>Gender</i>	<i>Work</i>	<i>Home</i>	<i>Other</i>
Male	60	20	10
Female	20	30	10

At the .01 level of significance, can we conclude that gender and the location of the accident are related?

<i>Observed frequencies (f_o)</i>				
<i>Gender</i>	<i>Work</i>	<i>Home</i>	<i>Other</i>	<i>Total</i>
Male	60	20	10	$r_1=90$
Female	20	30	10	$r_2=60$
Total	$c_1=80$	$c_2=50$	$c_3=20$	150

2 rows
 $r=2$

3 columns
 $c=3$

Observed cell frequency f_{12}

There are 6 cells as there are 2 rows and 3 columns.

<i>Expected frequencies ($\hat{E}_{ij}$)</i>				
<i>Gender</i>	<i>Work</i>	<i>Home</i>	<i>Other</i>	<i>Total</i>
Male	$\frac{(80)(90)}{150}$ $= 48$	$\frac{(50)(90)}{150}$ $= 30$	$\frac{(20)(90)}{150}$ $= 12$	90
Female	$\frac{(80)(60)}{150}$ $= 32$	?	?	60
Total	80	50	20	150

Step 1: H_0 : Gender and location are not related.

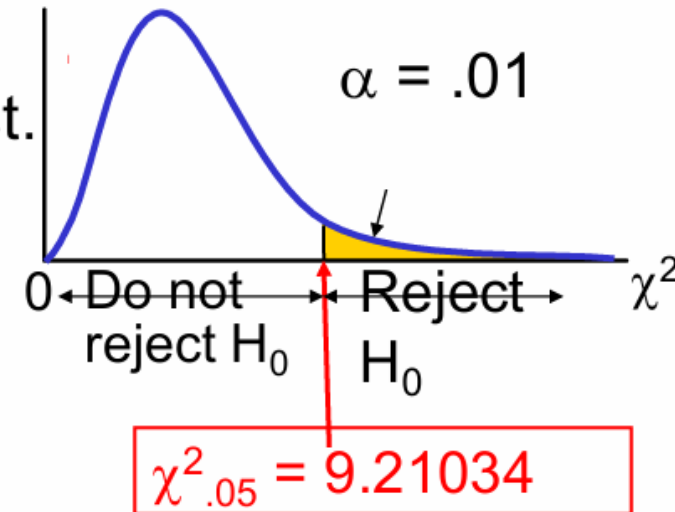
H_1 : Gender and location are related.

Step 2: The level of significance is set at .01.

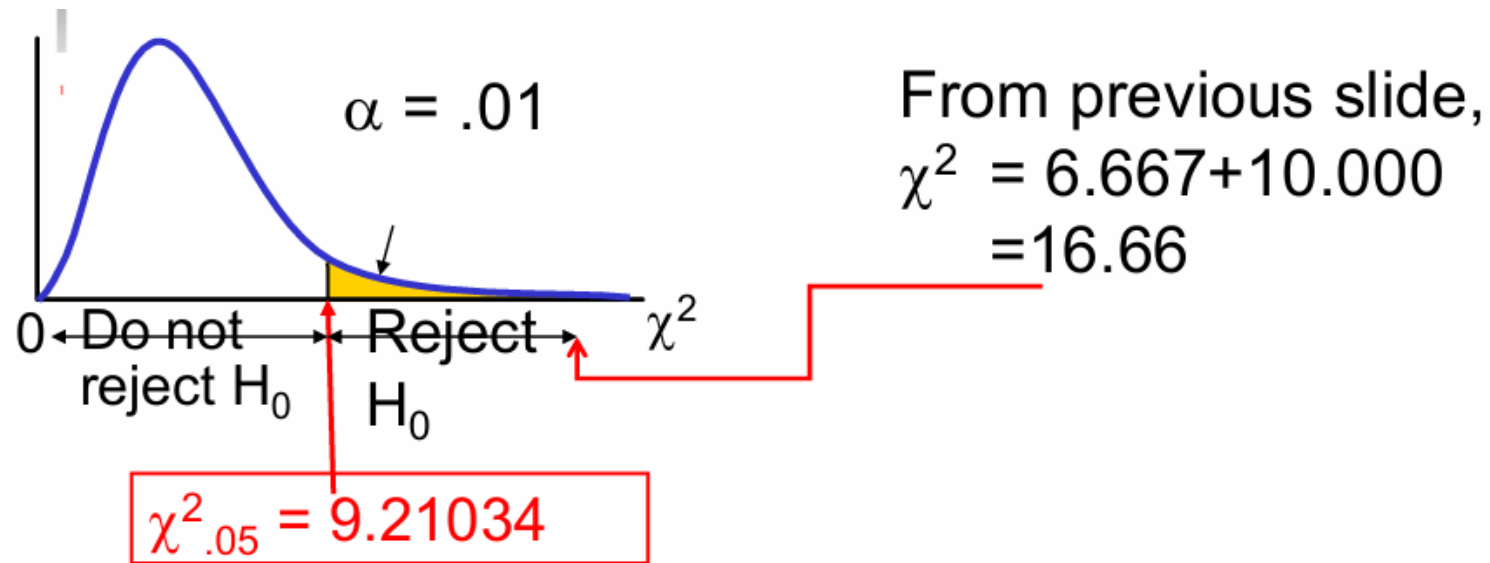
Step 3: The expected cell frequencies are
48,30,112,32,20,8 > 5.

We use chi-square test.

Step 4: $df = (r-1)(c-1)$
 $= (2-1)(3-1) = 2$.
The critical χ^2
is 9.21034.



$\chi^2: (f_{ij} - \hat{E}_{ij})^2 / \hat{E}_{ij}$				
<i>Gender</i>	<i>Work</i>	<i>Home</i>	<i>Other</i>	<i>Total χ^2</i>
Male	$\frac{(60-48)^2}{48}$	$\frac{(20-30)^2}{30}$	$\frac{(10-12)^2}{12}$	6.667
Female	$\frac{(20-32)^2}{32}$	$\frac{(30-20)^2}{20}$	?	?
Total				?



Since $\chi^2 = 16.667 > 9.210232$, reject the null hypothesis and conclude that there is a relationship between the location of an accident and the gender of the person involved.